📘 AI/ML Internship – Day 55 Notes & Tasks

🟢 Module 7: TF-IDF (Term Frequency – Inverse Document Frequency)

📌 Part 1: Recap of Day 54

Yesterday we learned:

✅ Bag of Words (BoW)

✅ Vocabulary Creation

✅ Word Frequency Counting

✅ CountVectorizer

Today we will learn:

🔥 TF-IDF

A smarter version of Bag of Words.

📌 Part 2: Problem with Bag of Words

Suppose we have these documents:

Document 1:

AI is amazing

Document 2:

AI is powerful

Document 3:

AI is growing rapidly

Using BoW:

Word	Count
AI	1
is	1

Word	Count
amazing	1

Problem:

Words like:

AI
is
the
and

appear in almost every document.

BoW gives them high importance.

But are they really useful?

❌ No

📌 **Part 3: What is TF-IDF?**

Definition

TF-IDF is a feature extraction technique that measures how important a word is in a document relative to the entire dataset.

Simple Meaning

TF-IDF gives:

✅ Higher weight to important words

✅ Lower weight to common words

📌 **Part 4: Full Form**

TF

Term Frequency

How many times a word appears in a document.

IDF

Inverse Document Frequency

Measures how rare a word is across all documents.

TF-IDF

TF × IDF

📌 Part 5: Understanding Term Frequency (TF)

Formula:

TF =
(Number of times word appears)
/
(Total words in document)

Example:

Document:

AI AI AI is amazing

Total words:

5

AI appears:

3 times

TF:

Meaning:

AI occupies 60% of the document.

📌 Part 6: Understanding IDF

Suppose:

Dataset contains:

100 Documents

Word:

AI

appears in:

90 Documents

Word:

Quantum

appears in:

5 Documents

Which word is more unique?

✅ Quantum

IDF gives more importance to rare words.

📌 **Part 7: IDF Formula**

Interpretation:

Common Word

AI

Appears everywhere.

Low IDF.

Rare Word

Quantum

Appears rarely.

High IDF.

📌 Part 8: Real-Life Example

Imagine a classroom.

Every student says:

Good Morning

Common phrase.

Not special.

One student says:

Quantum Computing

Rare phrase.

Gets more attention.

TF-IDF works similarly.

📌 Part 9: TF-IDF Workflow

Raw Text
↓
Cleaning
↓
Tokenization
↓
Stop Word Removal
↓
TF Calculation
↓
IDF Calculation
↓
TF-IDF Score

↓
Feature Vector

📌 **Part 10: Example Dataset**

Document 1:

AI is amazing

Document 2:

AI is powerful

Document 3:

Machine learning is amazing

Vocabulary:

AI
is
amazing
powerful
Machine
learning

TF-IDF assigns:

Word	**Importance**
is	Low
AI	Medium
amazing	Medium
powerful	High
learning	High

📌 **Part 11: Why TF-IDF is Better than BoW**

BoW

Counts words only.

Example:

the
the
the
the

High count.

High importance.

TF-IDF

Realizes:

the

appears everywhere.

Assigns lower weight.

Result:

More meaningful features.

📌 Part 12: TF-IDF Using Python

Install Scikit-Learn:

```
pip install scikit-learn
```

Code:

```
from sklearn.feature_extraction.text import TfidfVectorizer

documents = [
   "AI is amazing",
   "AI is powerful",
   "Machine learning is amazing"
]
```

vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(documents)

print(X.toarray())

📌 Part 13: View Vocabulary

```
print(
   vectorizer.get_feature_names_out()
)
```

Output:

```
['ai',
 'amazing',
 'is',
 'learning',
 'machine',
 'powerful']
```

📌 Part 14: Understanding TF-IDF Scores

Example:

Word	Score
AI	0.45
amazing	0.62
is	0.12

Interpretation:

Higher score:

More Important

Lower score:

Less Important

📌 Part 15: Real-World Applications

Search Engines

Google Search

Chatbots

Understanding important keywords.

Document Classification

Categorizing documents.

Spam Detection

Spam vs Non-Spam.

Resume Screening

Finding important skills.

📌 Part 16: Example – Spam Detection

Email:

Congratulations!
You won a lottery.
Claim prize now.

Important words:

lottery
prize
claim

TF-IDF assigns higher weights.

Machine detects spam easily.

📌 Part 17: BoW vs TF-IDF

Bag of Words	TF-IDF
Counts words	Calculates importance
Simple	Smarter
Less accurate	More accurate
Gives equal importance	Prioritizes meaningful words
Beginner Friendly	Industry Preferred

📌 Part 18: Interview Questions

What is TF-IDF?

A feature extraction technique that measures the importance of words.

Full form of TF-IDF?

Term Frequency – Inverse Document Frequency.

Why use TF-IDF?

To reduce the impact of common words and highlight important words.

Difference between TF and IDF?

TF measures frequency inside a document.

IDF measures rarity across documents.

Why is TF-IDF better than BoW?

It considers word importance, not just frequency.

📌 Part 19: Theory Questions

1. What is TF-IDF?

2. Define TF.
3. Define IDF.
4. Why is IDF important?
5. Compare BoW and TF-IDF.

🎯 Tasks for Day 55

💻 Practical Task 1

Install:

pip install scikit-learn

💻 Practical Task 2

Create TF-IDF vectors for:

AI is amazing
AI is powerful
Machine learning is amazing

💻 Practical Task 3

Print vocabulary using:

get_feature_names_out()

💻 Practical Task 4

Compare BoW and TF-IDF outputs.

💻 Practical Task 5

Identify the most important words using TF-IDF scores.